



Figure 2: Impact of vaccination and quarantine on the infected populations

The results are shown in Figure 2. The infections now peak at 4 per cent about 70 days after the outbreak. The effect of a quarantine is dramatic even with only a third of the infected population isolated.

The intention of the above examples is to show the ease with which various models can be created and to analyse the impact of certain safety measures. Strategies can be implemented and monitored. The effort is not in coding. It is in developing models that can be verified

by observations. The model can be enhanced to include births, deaths and disease-induced mortalities and much more, as shown in the second reference.

The concept of viruses has been applied to ideas (meme), computer malware, viral marketing, etc. An interesting example is an application to the modelling of a zombie attack www.wired.com/wiredscience/2009/08/zombies and www.mathstat.uottawa.ca/~rsmith/Zombies.pdf.

With the Scipy *integrate* module, you are not just restricted to understanding the dynamics of viruses and their impact on humans. Another very important model in biological systems is the predator-prey model. You can find a tutorial on using the Scipy *integrate* module to solve those equations at www.scipy.org/LotkaVolterraTutorial.

Furthermore, *pyplot* offers tremendous versatility and flexibility in presenting the results in an attention-grabbing way that quickly conveys the desired information. **END** 

References for the SIR model

- www.math.duke.edu/education/ccp/materials/diffcalc/sir/sir3.html
- wiki.deductivethinking.com/wiki/Epidemiology

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